THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** $2\,Mg + O_2 \longrightarrow 2\,MgO$	Two Mg atoms combine with one O2 molecule; MgO is the formula of magnesium oxide.
2	**(b)** 10	Bases turn red litmus blue and have pH above 7.
3	**(b)** $Fe + CuSO_4$ solution	Fe is more reactive than Cu; others involve a less reactive metal.
4	**(b)** 4	O=C=O has two double bonds: 4 shared pairs.
5	**(c)** excretion	Kidneys filter blood and form urine.
6	**(c)** dendrite	Dendrite tip receives the signal.
7	**(b)** at a definite plane related to the whip-like flagellum	Leishmania has a definite orientation of division; Amoeba divides in any plane.
8	**(b)** Gregor Johann Mendel	Mendel worked on garden pea plants.
9	**(c)** second law of reflection	Statement of the second law.
10	**(c)** retina	Light-sensitive screen at the back.
11	**(c)** coulomb	1 C is the charge of about 6×10^{18} electrons.
12	**(b)** from the north pole to the south pole	By convention, outside the magnet.
13	**(c)** biotic and abiotic components	All living and non-living parts interact.
14	**(d)** $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$	Ions are exchanged between the two compounds.
15	**(d)** 1.4	Stomach has hydrochloric acid.
16	**(d)** its easy loss of an electron	Sodium readily forms Na^+.
17	**(b)** long-chain carboxylic acids	Fatty acid salts.
18	**(c)** CO_2, water and energy	Glucose is fully oxidised.
19	**(d)** phototropism	Positive phototropism.
20	**(c)** pollen grain	Pollen grains carry the male gametes.
21	**(c)** all round yellow	Round and yellow are dominant.
22	**(d)** in vacuum to that in the medium	n = c/v.
23	**(a)** a concave lens	The concave lens diverges the rays.
24	**(d)** directly proportional to the potential difference, at constant temperature	V = IR.
25	**(d)** a bar magnet	One end behaves as N and the other as S.
26	**(c)** producers	Green plants fix energy.
27	**(a)** zinc is more reactive and displaces copper from its salt	Reactive metal displaces less reactive metal.
28	**(a)** magnesium hydroxide	$Mg(OH)_2$ neutralises acid.
29	**(d)** electrolysis of molten chlorides	They are too reactive for carbon reduction.
30	**(a)** catenation	Catenation.
31	**(b)** nephron	Each kidney has about a million nephrons.
32	**(b)** adrenaline	Adrenal glands.
33	**(c)** testes	Testes are the male sex organs.
34	**(a)** the type of sperm that fertilises the egg	Father's sperm may carry X or Y.
35	**(d)** a converging lens	It converges parallel rays.
36	**(a)** dispersion	Newton's prism experiment.
37	**(c)** R1 + R2 + R3	Rs = R1 + R2 + R3.
38	**(b)** induced current in a coil due to a changing magnetic field	It was discovered by Faraday.

39	**(d)** stratosphere	In the upper atmosphere.
40	**(d)** Displacement	More reactive Al displaces Fe from its oxide.
41	**(a)** HCl	Egg-shell is $\mathrm{CaCO_3}$; acids release $\mathrm{CO_2}$.
42	**(c)** applying grease	Paint/zinc cannot be used on a pan used for cooking; grease is suitable.
43	**(b)** 16	3n + 1 = 16.
44	**(d)** transport of water	Xylem carries water and minerals.
45	**(d)** synapse	Chemical signal crosses it.
46	**(d)** Plasmodium	Plasmodium divides into many daughter cells at once.
47	**(a)** it has many contrasting characters and is easy to grow	Pea has clear contrasting traits and short life cycle.
48	**(c)** virtual, erect and of the same size as the object	Plane mirror image is virtual, erect, same size, laterally inverted.
49	**(b)** outer surface of the cornea	The cornea has the greatest change in refractive index.
50	**(d)** ampere	1 A = 1 C/s.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(b)** the south pole to the north pole	They form closed loops.
2	**(b)** Temperature	Non-living factor.
3	**(b)** $BaSO_4$	Barium sulphate is insoluble in water.
4	**(b)** 5.5	Enamel dissolves below pH 5.5.
5	**(d)** calcium	Ca reacts less violently and floats.
6	**(b)** formation of micelles	Hydrophobic tails trap oil in micelles.
7	**(d)** lactic acid	Lactic acid causes cramps.
8	**(c)** gravity	Positive geotropism.
9	**(b)** ovary	After fertilisation the ovary ripens into fruit.
10	**(c)** 9 : 3 : 3 : 1	Four phenotypes appear.
11	**(b)** bends towards the normal	Glass is optically denser.
12	**(b)** a convex lens	A convex lens converges the rays.
13	**(a)** ohm	1 ohm = 1 V/A.
14	**(b)** makes an electromagnet	The core becomes magnetised.
15	**(a)** Grass, grasshopper, frog, snake	Producer to consumers in order.
16	**(b)** 2	$H_2 + Cl_2 \longrightarrow 2\,HCl$
17	**(a)** basic solution	Pink in bases, colourless in acids.
18	**(a)** impure copper	Impure copper dissolves.
19	**(d)** four valence electrons	Difficult to gain or lose 4 electrons.
20	**(c)** urinary bladder	Bladder.
21	**(c)** pituitary	It controls other glands.
22	**(d)** sperm formation needs a temperature lower than body temperature	A lower temperature is essential for sperm production.
23	**(d)** XX	Both are X.
24	**(a)** +2 D	P = 1/0.5 m = 2 D.
25	**(d)** Red	Red has the longest wavelength.
26	**(c)** current	Same current.
27	**(b)** Michael Faraday	Faraday, 1831.
28	**(a)** three oxygen atoms	O3.
29	**(d)** Calcium oxide (quick lime)	Quick lime reacts with water to give slaked lime used for whitewashing.
30	**(b)** 16 mL	Proportion: 8 x 20/10 = 16 mL.
31	**(c)** calcium	A metal forms an ionic oxide; calcium oxide is ionic and dissolves.
32	**(d)** C_2H_4	Ethene has a double bond.
33	**(b)** carbon dioxide, water, chlorophyll and sunlight	Photosynthesis needs all of these.
34	**(a)** insulin	Insulin is an animal hormone.
35	**(d)** budding	A bud grows out as an outgrowth and detaches.
36	**(c)** heredity	Definition of heredity.
37	**(d)** pole	The pole lies on the mirror.
38	**(d)** iris	Iris adjusts the amount of light.
39	**(a)** ammeter connected in series	An ammeter has a very low resistance.
40	**(d)** never intersect each other	Two directions at a point are impossible.
41	**(d)** Green plants	They make food by photosynthesis.

42	**(b)** gain of oxygen or loss of hydrogen	Definition of oxidation (in terms of oxygen/hydrogen).
43	**(a)** tartaric acid	Tamarind contains tartaric acid.
44	**(b)** copper	Copper is below hydrogen in the series.
45	**(a)** calcium and magnesium salts forming scum	Insoluble Ca/Mg salts of fatty acids.
46	**(a)** ethanol and carbon dioxide	Fermentation.
47	**(a)** thigmotropism	Response to touch.
48	**(b)** seed	The fertilised ovule becomes the seed.
49	**(b)** 3	The 3 round green seeds are R_yy.
50	**(a)** bends away from the normal	Speed increases.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** in front of the retina	Eyeball is too long or lens too powerful.
2	**(b)** a straight line through the origin	Slope = R.
3	**(b)** soft iron wrapped with a coil carrying current	It is a temporary magnet.
4	**(c)** trophic levels	Each is a feeding level.
5	**(a)** 2 : 1	$2\,H_2O \longrightarrow 2\,H_2 + O_2$.
6	**(c)** red	Red in acid, yellow in base.
7	**(d)** acidified copper sulphate solution	Cu^{2+} ions move to the cathode.
8	**(b)** 2	n-butane and isobutane.
9	**(b)** glomerulus	Glomerulus in Bowman's capsule.
10	**(c)** growth hormone	Pituitary.
11	**(c)** ovaries	Ovaries produce eggs.
12	**(a)** XY	One X and one Y.
13	**(a)** dioptre	1 D = 1 m^-1.
14	**(c)** Violet	Shortest wavelength.
15	**(c)** potential difference	Same voltage.
16	**(b)** a magnet is moved towards or away from it	Relative motion changes the flux.
17	**(a)** harmful ultraviolet radiation	UV rays can cause skin cancer.
18	**(b)** remove the protective oxide layer from its surface	The oxide coating prevents the metal from burning.
19	**(d)** antacid	Antacids neutralise excess stomach acid.
20	**(b)** zinc is more reactive than tin	Zinc would react with acids in food.
21	**(b)** C_4H_8	Homologues differ by CH_2.
22	**(b)** mitochondria	Aerobic respiration in mitochondria.
23	**(b)** auxin	Auxin elongates cells on the shaded side.
24	**(a)** spores	Sporangia release many spores.
25	**(a)** all tall	Tallness is dominant.
26	**(a)** R/2	f = R/2.
27	**(d)** accommodation	Done by ciliary muscles.
28	**(d)** voltmeter connected in parallel	A voltmeter has a very high resistance.
29	**(a)** the closeness of the field lines	Crowded lines mean a strong field.
30	**(c)** consumers	Heterotrophs.
31	**(d)** H_2	H_2 gains oxygen to form H_2O.
32	**(c)** methanoic acid	Methanoic acid causes the burning pain.
33	**(b)** $MgCl_2$	Metal + non-metal compound is ionic.
34	**(c)** an alcohol	Esterification.
35	**(a)** ATP	ATP.
36	**(c)** chemotropism	Chemical stimulus.
37	**(c)** pollination	This is the definition of pollination.
38	**(d)** traits are inherited independently	Independent assortment.
39	**(d)** 2×10^8 m/s	v = c/n.
40	**(a)** farther than 25 cm	A person cannot see nearby objects clearly.
41	**(c)** directly proportional to its length	R is proportional to L.

42	**(b)** increasing the current or the number of turns	More turns, more current.
43	**(b)** producers	Plants and algae.
44	**(d)** thermal decomposition	Heat breaks down limestone.
45	**(a)** 7	Water is neutral.
46	**(b)** copper and zinc	Brass: Cu + Zn.
47	**(c)** non-conductor and very hard	Each carbon bonded to four others.
48	**(a)** dialysis	Haemodialysis.
49	**(a)** insulin	Blood sugar is high.
50	**(c)** oviduct (fallopian tube)	The egg meets the sperm in the oviduct.

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